

$$\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda;z) = \int_0^1 dr_1 \int_0^1 dr_2\; r_2\, e^{-\lambda_{1,+\dot{a}}[r_1 z_{12}+(1-r_1)z_{14}]^{+\dot{a}}} \times \lambda_{1,+}{}^{\dot{b}} \frac{\partial}{\partial z_{24}^{\text{red},+\dot{b}}} \mathcal{I}_{\triangle}^{\text{raw cut}}\Big(\lambda_2 + r_1\lambda_1, \lambda_3, \lambda_4 + (1-r_1)\lambda_1; z^{\text{red}}\Big).$$

$$z_{24}^{\text{red}} = r_2 z_{24} + (1-r_2)(z_{14}-z_{12}), \qquad z_{23}^{\text{red}} = z_{23}, \quad z_{34}^{\text{red}} = z_{34}.$$